

EXPERIMENT: Write a Prolog program to implement Monkey Banana Problem

Aim

To write a Prolog program to solve the Monkey Banana Problem using logical reasoning and state-space search, where a monkey must move a box to reach the bananas and obtain them.

Objectives

- To represent the different states of the monkey, box, and bananas.
- To use Prolog rules to describe possible actions.
- To find a sequence of actions that allows the monkey to get the bananas.
- To understand problem-solving using state-space representation.

Algorithm

1. Start with the initial state.
2. Check whether the monkey is already on the box below the bananas.
3. If not, move the monkey toward the box.
4. Push the box to the banana location.
5. Climb onto the box.
6. Grab the bananas.
7. Display the sequence of actions.
8. Stop when the bananas are obtained.

CODE

```
monkey_position = "door"
```

```
box_position = "window"
```

```
banana_position = "middle"
```

```
monkey_status = "floor"
```

```
has_banana = False
```

```
print("=" * 50)

print("    MONKEY BANANA PROBLEM")

print("=" * 50)


print("\nInitial State:")
print("Monkey position :", monkey_position)
print("Box position   :", box_position)
print("Banana position :", banana_position)
print("Monkey status  :", monkey_status)
print("Has banana     :", has_banana)
print("\nSteps to get the bananas:")
print("-" * 50)

monkey_position = box_position
print("1. Monkey moves to the box.")

box_position = banana_position
monkey_position = banana_position
print("2. Monkey pushes the box under the bananas.")

monkey_status = "onbox"
print("3. Monkey climbs onto the box.")

if monkey_position == banana_position and monkey_status == "onbox":
    has_banana = True
print("4. Monkey grabs the bananas.")

print("-" * 50)

print("\nFinal State:")
print("Monkey position :", monkey_position)
print("Box position   :", box_position)
```

```
print("Banana position :", banana_position)
print("Monkey status  :", monkey_status)
print("Has banana    :", has_banana)
if has_banana:
    print("\nSUCCESS: Monkey got the bananas!")
else:
    print("\nFAILURE: Monkey could not get the bananas.")
print("=" * 50)
```

OUTPUT

```
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\shaik arshad basha\AppData\Local\Programs\Python\Python312\python.exe
=====
MONKEY BANANA PROBLEM
=====

Initial State:
Monkey position : door
Box position : window
Banana position : middle
Monkey status : floor
Has banana : False

Steps to get the bananas:
-----
1. Monkey moves to the box.
2. Monkey pushes the box under the bananas.
3. Monkey climbs onto the box.
4. Monkey grabs the bananas.
-----

Final State:
Monkey position : middle
Box position : middle
Banana position : middle
Monkey status : onbox
Has banana : True

SUCCESS: Monkey got the bananas!
=====
>>> |
```

Ln: 33 Col: 0

Result

The Monkey Banana Problem was successfully implemented in Prolog. The program finds a sequence of actions in which the monkey moves to the box, pushes it below the bananas, climbs onto it, and finally grabs the bananas.

Conclusion

The program demonstrates state-space problem solving in Prolog. By representing the monkey, box, banana position, and monkey's status as states, Prolog can determine the actions required to reach the goal state.